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Review

Nanobubble technology to enhance energy recovery from anaerobic digestion of organic solid wastes: Potential mechanisms and recent advancements

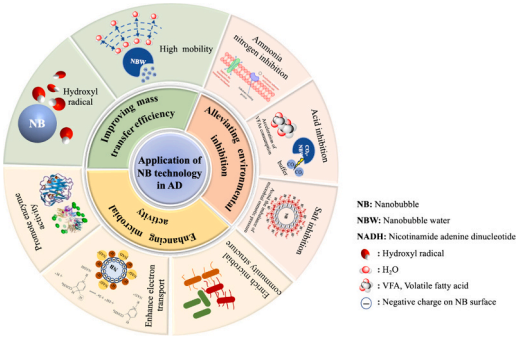
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HIGHLIGHTS

- •OH produced by the collapse of NB is beneficial for the oxidation of organic matter.
- NB could buffer adverse factors to alleviate environmental inhibition of AD reactors.
- NB could improve the enzymes activities and microbial diversity in the AD system.
- The high water mobility of NB could increase the nutrient transport efficiency.

GRAPHICAL ABSTRACT



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ABSTRACT

Nanobubble (NB) technology has gained popularity in the environmental field owing to its distinctive characteristics and ecological safety. More recently, the application of NB technology in anaerobic digestion (AD) systems has been proven to promote substrate degradation and boost the production of biogas (H_2 and/or CH_4). This review presents the recent advancements in the application of NB technology in AD systems. Meanwhile, it also sheds light on the underlying mechanisms of NB technology that contribute to the enhanced biogas production from AD of organic solid wastes. Specifically, the working principles of the NB generator are first summarized, and then the structure of the NB generator is optimized to accommodate the demand for NB

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characteristics in the AD system. Subsequently, it delves into a detailed discussion of how the addition of nanobubble water (NBW) affects AD performance and the different factors that NB can potentially contribute. As a simple and environmentally friendly additive, NBW was commonly used in the AD process to enhance the fluidity and mass transfer characteristics of digestate. Additionally, NB has the potential to enhance the functionality of different types of microbial enzymes that play crucial roles in the AD process. This includes boosting extracellular hydrolase activities, optimizing coenzyme F_{420}, and improving cellulase function. Finally, it is proposed that NBW has development potential for the pretreatment of substrate and inoculum, with future development being directed towards this aim.

1. Introduction

Anaerobic digestion (AD) serves as a crucial technology for the resourceful utilization of biomass energy, efficiently converting the inherent energy of organic solid waste into biogas for combustion or power generation, particularly when dealing with agricultural waste with an energy content of 16 MJ/kg (Rahmani et al., 2023). Nowadays, AD has been extensively applied to treat sewage sludge, livestock manure, municipal organic waste, and agricultural waste, facilitating the achievement of environmental protection, greenhouse gas emission reduction, and renewable energy production (Laiq et al., 2019). Nevertheless, the large-scale applications of AD technology are still hampered by several constraints. For instance, although the hydrolysis stage serves as a pivotal step in the AD process, it frequently exhibits sluggish kinetics when dealing with complex substrates, resulting in low hydrolysis efficiency. Consequently, the hydrolysis stage is widely recognized as the rate-limiting step in the AD process of lignocelluloses and high solid organic wastes (Tamang et al., 2023). On the other hand, methanogenesis is a multifactorial process involving the synergistic effects of multiple neutrophilic bacteria, so various factors (such as oxygen content, pH, temperature, organic load, etc.) can affect the methanation process (Li et al., 2019; Liu et al., 2021). Researchers are currently making efforts to enhance AD performance through pretreatment, additive supplementation (e.g., biological additives, micronutrients, carbon-based materials, and metal-based materials), and reactor optimization (Liu et al., 2021; Meegoda et al., 2018; Nguyen et al., 2021). However, difficulties in maintaining operating conditions and the addition of expensive chemicals are accompanied by increased costs together with subsequent treatment of some additives. For example, the external addition of micronutrients in large-scale AD plants, often relying on expensive and specialized chemicals, faces challenges due to the absence of standardized addition schemes, potentially reducing environmental benefits and increasing operating costs (Bardi et al., 2023). Additionally, biochar, as an environmentally friendly additive in the AD system, also faces unresolved issued because its effectiveness is closely related to the inherent properties (raw materials, synthesis methods, process parameters, etc.) (Nie et al., 2024). Therefore, novel strategies that increase methane production combined with cost and pollution reduction are urgently required.

Nanobubbles (NBs) refer to nanoscopic gaseous (typically composed of air, O_2, CO_2, N_2, and other gases) cavities within aqueous solutions, defined by the International Standard Organization (ISO) as having a volume-equivalent diameter below 1 μm and exhibiting the ability to alter the standard properties of aqueous solution (Jia et al., 2023). Most recently, NB technology has attracted more and more attention due to its unique properties and absence of secondary pollution, and researchers have explored its utilization in enhancing AD performance, leveraging its attributes such as prolonged residence time, high mass transfer efficiency, negative surface charge, and collapse to produce free radicals like •OH radicals (Atkinson et al., 2019; Khan et al., 2020).

Unlike coarse bubble (CB), NB is almost unaffected by the buoyancy in solution, performing free Brownian motion and rising slowly, thus enabling them to remain in solution for 2 weeks or longer (Lyu et al., 2019). Generally, the solubility of CB in water is always affected by the environmental pressure and limited by their saturation solubility, whereby it is difficult for CB to reach above saturation solubility under normal environmental conditions. In contrast, according to Henry's law, NB gradually shrinks during their slow ascent, and the surface tension of NB is increased with the increase in gas compression and internal pressure (Koshoridze and Levin, 2018; Lohse and Zhang, 2015). Therefore, the internal pressure of NB is higher than the external atmospheric pressure, which enables it to break the limitation of its solubility under the atmospheric pressure to obtain higher solubility (Craig, 2011). It has been proven that the gas transfer coefficient of NB can be increased by 11 times in comparison to CB under the same volume of gas transport conditions in water (Atkinson et al., 2019). Moreover, NB possess the characteristics of the characteristics of the negatively charged surface and high zeta potential primarily due to the adsorption of hydroxyl ions on the interface surface of NB, which is also one of the reasons why NB is more stable in solution. Negative surface charge and high zeta potential of NB are important factors affecting the mass transfer of cations in solution, hydrophobic properties, and water mobility, i.g., 1) efficiently control the mass transfer process of cations by attracting and fixing them through electrostatic interactions, thereby significantly boosting the absorption efficiency of local nutrients; 2) influence the hydrophobic properties by modifying the arrangement of surrounding water molecules, indirectly enhancing particle adhesion (Chuenchart et al., 2021). Additionally, NB can produce reactive species upon their collapse (Atkinson et al., 2019; Ghadimkhani et al., 2016). •OH radicals are the main reactive species continuously produced at the gas-liquid interface when NB collapses in aqueous solution, which could break the stable structure of complex substrates and accelerate their degradation (Agarwal et al., 2011; Atkinson et al., 2019).

Apart from the unique physical and chemical properties, NB also has non-toxic and environmentally friendly characteristics. Consequently, the application of NB technology in environmental protection, biomedicine, food processing, and other fields will not generate secondary pollution, which is in line with the sustainable development goals (Phan et al., 2020; Tekile et al., 2017; Xiong et al., 2021). Based on these properties, researchers have applied NB technology to improve energy recovery from AD systems recently, and it has been proven that nanobubble water (NBW) could promote substrate degradation, improve microbial activity, and increase hydrogen or/and methane production. The specific application examples are summarized in Table 1. The results show that N_2-NBW can contribute to the establishment of a strictly anaerobic environment and enhance cellulase activity with promoted lignocellulose degradation for methanogenesis (Wang et al., 2020a). In comparison, air-NBW and O_2-NBW can create a micro-oxygen environment, which appears to be antithetical to the strictly anaerobic environment required for an AD system. Interestingly, the addition of air-NBW or O_2-NBW in the AD system enhanced the methane production efficiency instead of inhibiting methanogenesis (Hou et al., 2022; Wang et al., 2020c). The prospective mechanisms are related to promoting the degradation of substrates, accelerating the generation and consumption of soluble nutrients, and improving the activity of facultative anaerobic bacteria (Fu et al., 2016; Zhu et al., 2022).

It can be observed that NBW of different gas types exhibits enormous potential in the AD process of organic wastes such as lignocellulosic biomass, food waste (FW), livestock manure, and so on (Fan et al.,

2021a; Hou et al., 2021a; Wang et al., 2020a). However, the interaction mechanisms between NBW and different substrates in AD are unclear, especially the enhancement of mass transfer properties of substrate by NBW. In addition, the generation methods and conditions of NB directly influence the characteristics (particle size, bubble concentration, zeta potential, etc.) of NB and hence impact the AD performance (Wang et al., 2019; Wang et al., 2021). It is also necessary to clearly understand the working principles of the NB generator to design and optimize its operation according to the requirements of AD systems for different organic solid waste treatment.

In this review, the working principles and characteristics of several NB generators were described, focused on the optimization for structure design of venturi-type NB generators. Meanwhile, the mechanisms associated with the enhancement of AD performance and the effects of NB on the mass transfer characteristics of digestate were discussed in the context of the advances of NB technology in AD of different organic solid waste systems. Finally, the potential of NBW for the pretreatment of substrates and inoculum and future perspectives were prospected.

2. NB generation and optimization for structure design of NB generator

2.1. NB generators

The properties of NB prepared by different generation mechanisms may vary, and several design parameters may also influence the particle size and concentration of NB. As a result, different fields apply NB with different properties due to different generation methods. At present, NB is mainly manufactured by the porous ceramic membrane, circulation extrusion, electrochemistry, and cavitation.

2.1.1. Porous ceramic membrane NB generator

As reported, the porous ceramic membrane NB generator mainly consists of a pressurized gas cylinder, gas regulator, gas flow meter, porous ceramic tube, filter, and connecting pipeline (Ahmed et al., 2018a). After compressed gas is introduced to the exterior of the porous membrane and passes through the pores, the liquid flowing inside the membrane is cut off faster and the bubbles that form on the membrane surface are separated without any shear force being applied. Ahmed et al. (2018b) provided a detailed report on the relationship between the NB properties generated by porous ceramic membrane filters and the surface coating, pore size, and injected gas pressure. Results show that the size and Zeta potential of NB were dependent on the pore size of the ceramic tubular membrane under the same injection pressure, and the higher the injected pressure would result in a more stable generation of NB. Unfortunately, another study suggested that the porous ceramic membrane method will require a longer time to generate stable NB, taking an average of 90 min.

2.1.2. Circulation extrusion NB generator

The circulation extrusion method is mainly pressure-driven, where a piston periodically alters the pressure inside the tube and the gas is repeatedly compressed into the water to produce NB. This method generates NB with a high negative Zeta potential (-40 ± 2 mV) and the concentration of NB is up to 1.92×10^{10} mL^{-1} after 600 repetitions of compression (Jin et al., 2019). Additionally, it's also possible to combine pressurized mechanical circulation and gas-liquid circulation by adjusting the appropriate gas-liquid flow rate and internal pressure to generate a strong shear force to break down CB into NB (Wang et al., 2019).

2.1.3. Electrochemically generated NB

NB was found to be formed in commercial electrochemical flow cells, and the principle is electrolyzed to oxygen and hydrogen through electrochemical action and then present in solution as NB with concentrations about 10^7–10^8 mL^{-1} and particle size in the range of 94–529 nm (Zhang and Seddon, 2016). Nevertheless, it must be considered that the electrochemical method needs to maintain a sufficient electrical potential to produce gas, requiring a large energy consumption that limits its application in practice. In general, the electrochemical method producing NB is mainly applicable for smaller-scale manufacturing and cannot be used for industrial-scale processes.

2.1.4. Cavitation generated NB

Agarwal et al. assumed that the formation, growth, and collapse of NB is a cavitation process seemingly like boiling with different driving mechanisms (Agarwal et al., 2011). When exposed to higher pressures, the cavities will implode and generate strong shock waves, resulting in the formation of NB. According to the cavitation process inducing bubble formation mediums are different, they can be broadly classified as ultrasonic, optical, particle, and hydrodynamic cavitation.

Ultrasonic cavitation caused by low pressure generated via high-frequency sound waves for achieving cavitation effectiveness. Nirmalkar et al. (2019) tested that the NB concentrations were up to 2×10^8–1×10^9 mL^{-1} and NB sizes of 100–120 nm by using a 20 kHz probe-type ultrasonic processor to generate NB in surfactant solutions. However, the amount of dissolved gas had a significant effect on the concentration of NB generated by ultrasonic cavitation. For instance, when the dissolved gas was partially vacuumed, the concentration of NB is about 4×10^7 mL^{-1} with one order of magnitude reduction, indicating that

Table 1
Study on the effect of NB on AD performance.

Substrates	Temperature and initial pH	Gas type	Max methane production (mL/g-VS or mL/g-TOC)	Methane production (%, compared with DW)	References
Corn straw	35 °C 7.0	N_2/O_2	127.3/141.7	10 %/22 %	(Wang et al., 2020a)
Cellulose	35 °C 8.0	N_2/Air/O_2	196.1/193.3/233.0	9 %/10 %/30 %	(Wang et al., 2020c)
Cellulose	35 °C 7.0	Air/CO_2	264.0/246.0	18 %/10 %	(Wang et al., 2020b)
Waste active sludge (WAS)	36 °C 7.0	CO_2/H_2/Air/N_2	387.8/372.8/369.5/361.2	28 %/23 %/23 %/19 %	(Wang et al., 2019)
Food waste	38 °C 7.5/8.0/9.0	Air	297.2	25 %	(Hou et al., 2022)
Food waste	38 °C 7.5	Air	322.2	43 %	(Hou et al., 2021a)
Food waste	38 °C 7.5	N_2/Air	347.6/373.6	16 %/24 %	(Hou et al., 2021b)
Pig manure	35 °C 5.5/6.0/6.5	Air	257.0	17 %	(Fan et al., 2021a)
Pig manure	35 °C 7.0	CO_2	230.0	25 %	(Fan et al., 2020b)
Pig manure	35 °C 7.0	Air/H_2/CO_2	192.0/195.0/225.0	19 %/21 %/39 %	(Fan et al., 2020a)
Alkaline lignin +WAS	37 °C 7.0	N_2	443.1	17 %	(Yang et al., 2020)
WAS	37 °C 7.0	N_2	402.1	29 %	(Yang et al., 2019)
Acetic acid + Lignin	35 °C 7.0	N_2/CO_2	728.0/703.0	22 %/17 %	(Guengerich and Yoshimoto, 2018)
Corn straw	35 °C 7.0	H_2	254.4	25 %	(He et al., 2022)
Corn straw	35 °C 7.0	CO_2	132.2	17 %	(Song et al., 2023)

adequate gas supply is essential for the preparation of NB by ultrasonic cavitation (Jadhav and Barigou, 2020). Concomitantly, ultrasonic cavitation has high power consumption, which is not conducive to practical applications.

In contrast to ultrasonic and hydrodynamic cavitation, optical and particle cavitation cannot lead to physical and chemical changes in the solution (Agarwal et al., 2011). The most common method for generating optical cavitation is to focus a short-pulse laser in a transparent liquid medium and high-speed particles passing through the liquid can also produce cavitation to form NB. Padilla-Martinez et al. adopted a novel optical cavitation method by using a low-power continuous laser incident on the highly absorbing liquid (Padilla-Martinez et al., 2014). Afterward, the absorbed light rapidly heated the liquid beyond its boiling in microseconds to milliseconds, resulting in the explosive liquid-vapor phase transition once the liquid was heated to its spin-nodal limit (about 300 °C for pure water). Compared to liquids, the volume of superheated vapor was increased by multiple orders of magnitude, thereby creating rapidly expanding vapor bubbles that eventually burst into smaller NB.

Hydrodynamic cavitation refers to the millions of cavities that could be formed when the throttling pressure drops sufficiently below the cavitation threshold pressure. The concentration of NB produced by hydrodynamic cavitation could be increased by one order of magnitude compared to ultrasonic cavitation. Furthermore, hydrodynamic cavitation is the most energy-efficient and cost-effective method for generating NB at experimental and industrial levels in comparison to the three above-mentioned cavitation methods (Azevedo et al., 2019). At present, the most common typical hydrodynamic cavitation NB generators include spiral liquid flow type, pressurized dissolution type, ejector type, venturi type, and so on.

2.2. Parameters optimization of venturi-type NB generator

In recent years, venturi-type NB generator has received more and more attention because of their simple structure, convenient installation without internal moving parts, less maintenance, high reliability, and promising applications across diverse fields like environmental purification, fish farming, agricultural production, and bioreactors (Kayaalp and Ozturkmen, 2016; Mitra et al., 2016; Zhao et al., 2019). Notably, the compact and uncomplicated design of venturi-type NB generators facilitates integration into existing AD systems, a crucial aspect for practical applications where spatial and installation constraints are prevalent. The primary working principle of the venturi-type NB generator is that the liquid enters the throat of the venturi at a certain speed driven by the circulation pump, and the inhaled gas is violently mixed with the liquid following the sudden change of liquid speed and pressure. In the negative pressure mixing chamber, a supersaturated gas-liquid mixture is formed, and then cavitation and nucleation occur. After several cycles of the above process, a liquid solution containing NB will be formed. Therefore, the gas-liquid flow rate and the design structure size have an important influence on the effectiveness of NB generation for venturi-type NB generators. Additionally, the application of NB technology to AD systems involves the generation of bubbles of the smallest possible size to be used throughout the AD cycle. If the operating conditions and size parameters of the Venturi-type NB generator are designed reasonably, NB may be prepared to meet the requirements of AD.

2.2.1. Effect of gas-liquid flow rate on the characteristics of NB

An investigation of the factors (including air, water, Reynolds number, and volume fraction) affecting the size distribution of bubbles in a venturi NB generator revealed that the size of the bubbles was inversely proportional to the liquid flow rate and increased with increasing volume fraction (Gordiychuk et al., 2016). Moreover, increasing the liquid flow rate could lead to a more uniform bubble size distribution. However, increasing the gas flow rate has the opposite effect, with the average diameter of the bubbles increasing as the gas flow rate increases at a fixed water flow rate. Consequently, it is necessary to strictly control the gas flow in addition to providing a strongly turbulent field with sufficient liquid flow to generate a reasonable bubble size to meet the application requirements (Sun et al., 2017).

Besides, bubble rupture in turbulent flows is primarily caused by dynamic pressure or shear stress on the bubble surface. This mechanism could be quantified by the Weber number. When the maximum hydrodynamic force of the liquid is higher than the surface tension, it will rupture into smaller bubbles. Here, the W_e in turbulent flow can be defined as Eq. (1)

$$W_e = F_i/F_\sigma \tag{1}$$

where F_i denotes the inertial force on the bubble and F_σ denotes the surface tension, which can be expressed as Eq. (2)

$$F_\sigma = \pi\sigma d \tag{2}$$

where d is the bubble diameter, calculated according to Eq. (3)

$$F_i \propto \frac{\rho\pi d^3}{6g}\left(\frac{\varepsilon_v C}{\rho}\right)^{2/3} \Big/ d^{1/3} \tag{3}$$

For a liquid flow in a pipe, the energy dissipation rate ε_v per unit volume can be expressed as Eq. (4)

$$\varepsilon_v = \left(\frac{0.316}{2g}\right)\frac{\mu^3}{\rho^2 D_t^4} R_e^{11/4} \tag{4}$$

where R_e is the Reynolds number of the bubble generator throat, which can be expressed as Eq. (5)

$$R_e = \frac{\mu d}{v} \tag{5}$$

where μ is the fluid dynamic viscosity and v is the fluid kinematic viscosity. Therefore, the above equations can be combined to obtain Eq. (6)

$$\frac{d}{D} \propto \left(\frac{g\sigma\rho D_t}{\mu^2}\right)^{3/5} R_e^{-1.1} \tag{6}$$

This indicates that the bubble size is proportional to the −1.1 power of *Re* in the throat of the NB generator, and the increase in *Re* could enhance the degree of bubble fragmentation. According to the results of Zhao et al. (2019), it can be concluded that the rupture process became more intense as the water flow rate increased, and the large bubbles were continuously ruptured, and numerous smaller bubbles were produced. This was attributed to the increase in the working power of the circulating pump prompting an increase in the inlet water flow rate and a consequent increase in the turbulence intensity and pressure drop. Therefore, it is possible to adjust the operating power of the circulating pump as necessary to obtain the desired size of bubbles.

2.2.2. Effect of generator geometric parameters on the characteristics of NB

In the venturi-type NB generator, the parameters design of convergent angle α, divergent angle β, throat length L, throat diameter d, outlet diameter D_o, and inlet diameter D_i have a significant impact on the performance of the generator. Reichmann et al. (2017) investigated the effects of β, L, d, and D_o on the generated bubble size. Results showed that (1) the bubble size was decreased with the increase of L and β; (2) the decrease of d and D_o was beneficial for generating smaller size bubbles; (3) the effect of α and inlet diameter on bubble size domesticated was not significant. These indicated that d and β play a key role in the performance of the bubble generator. Zhao et al. also derived similar conclusions by testing three β angles (7.5°, 10°, and 12.5°), where a larger divergence angle would lead to a stronger interaction between the gas and the liquid, and thus would produce smaller bubbles (Zhao et al.,

2019).

In summary, with the gradual reduction of d, the flow of the liquid in the pipe will increase the differential pressure between the front and back of the pipe, resulting in a corresponding increase in the acceleration and velocity of the fluid through the pipe. Consequently, this augmentation in flow dynamics elevates the R_e at the pipeline's rear, thereby amplifying the turbulence effect. Additionally, the position of the bubble bursting will also be advanced with the increase in β, and the increase in velocity and pressure changes will lead to an increase in the degree of bubble bursting. Therefore, the efficiency of the venturi-type NB generator can be significantly improved by appropriately reducing d and increasing β.

3. Mechanisms of NBW as an additive to enhance AD performance

3.1. Generation of reactive oxygen species

Reactive oxygen species (ROS) mostly include superoxide anion (O_2^-), hydrogen peroxide (H_2O_2), hydroxyl radicals (•OH), ozone (O^3), and singlet oxygen ($^1O_2$), which have high chemical reactivity due to the presence of unpaired electrons (Apel and Hirt, 2004; Trachootham et al., 2008). ROS can be divided into exogenous ROS and endogenous ROS. Specifically, exogenous ROS are mainly induced by chemical substances (H_2O_2, etc.) and exogenous radiation (X-ray, γ-rays, etc.), while endogenous ROS primarily originate from mitochondrial oxidants, nicotinamide adenine dinucleotide phosphate (NADPH), and so on (Finkel, 2011; Hamanaka and Chandel, 2010; Liu et al., 2016b). Normally, the strongly oxidizing properties of ROS would be regarded as reactive substances with certain damage to the organism because excessive amounts of reactive oxygen species can damage cells and produce pathological effects. On the contrary, it has also been reported that moderate amounts of ROS have some positive effects on cells (Kranner et al., 2010; Liu et al., 2016a).

Interestingly, NB has been proven to produce ROS without the addition of any chemicals. A large amount of charged ions accumulated on the surface of the NB will release chemical energy, accompanied by the generation of free radicals (Smirnov et al., 2018; Takahashi et al., 2007). Liu et al. (2016b) distinguished four types of ROS ($^1O_2$, H_2O_2, •OH, and O_2^-) in NBW and found that •OH radicals were the special ROS produced by NBW, the concentration of •OH radicals was positively correlated with the concentration of NB. •OH radicals possess strong oxidizing properties that assist in the oxidative of complex organic matter and promote the decomposition of substrates. In the AD system, complex substrates are generally difficult to completely degrade by microorganisms, whereas the •OH radicals generated by NB could contribute to the degradation of complex substrates, as shown in Fig. 1.

In addition, Yang et al. (2020) speculated that •OH radicals generated by NB may play an important role in the lignin depolymerization process with a 10 % increase in the lignin degradation rate. Another study also mentioned that the ability of NB to generate •OH radicals could improve the degradation rate of cellulose and hemicellulose, where O_2-NB were more likely to produce •OH radicals compared to N_2-NB (Wang et al., 2020a). However, it is important to note that although •OH radicals are instrumental in substrate degradation, excessive amounts may have negative effects on microorganisms in the AD system. A study by He et al. (2022) found that 60 % H_2-NBW was able to increase methane production by about 25 % and cellulose and hemicellulose degradation by about 33 % and 26 %, respectively. The addition of excess NBW may have an inhibitory effect on microbial growth and metabolism. Therefore, further research remains necessary to determine the optimal addition of NBW and to reduce the possible negative effects.

3.2. Alleviation of environmental inhibition

AD technology in the treatment of organic solid wastes is generally associated with high organic load, high ammonia nitrogen, severe acidification, and toxic inflow. As a result, microorganisms would be inhibited by these adverse environmental factors, resulting in a reduction in methane production (Laiq et al., 2019). Recent studies have shown that the addition of NBW could alleviate the inhibition of adverse factors on AD performance (summarized in Fig. 2).

As known, the AD process involves four main stages, including hydrolysis, acidification, acetogenesis, and methanogenesis. Hydrolysis/acidification in the early stage is always accompanied by the accumulation of VFAs with the consequent continuous drop in digestate pH. When the pH is lower than 6.5, microbial activity is inhibited because of the strict environmental requirements of methanogens (Li et al., 2017). Fan et al. (2021a)investigated the effects of air-NBW addition on the stability of the AD process by measuring the total alkalinity (TA), partial alkalinity (PA), and VFA/TA ratio. It was revealed that the addition of air-NBW could improve the TA and PA and decrease the VFA/TA ratio to maintain the stability of the AD process. In another study on the AD of high-solids swine manure (total solids (TS) = 8 %), the inhibition of the methanogenesis stage due to the large accumulation of VFA was relieved after the NBW addition, and reactor stability was gradually improved (Fan et al., 2020b). These results demonstrate that the addition of air-NBW could effectively alleviate acid inhibition and thus improve the stability of the AD system. This may be associated with a significant increase in methanogenic bacterial activity with the addition of NBW (Hou et al., 2021b). However, the in-depth mechanism associated with the addition of NBW to alleviate acid inhibition is uncertain.

High salinity and high ammonia nitrogen are also the main factors that inhibit methane production from the AD process of specific organic solid wastes, such as FW and animal manure. Different salt concentrations cause varying degrees of inhibition of hydrogen and methane production. Hou et al. (2021a) attempted to mitigate the adverse effects of high salinity on hydrogen and methane production from the two-stage AD of FW by adding air-NBW. As expected, the observed increase in hydrogen and methane yields by 21 %–65 % and 14 %–43 %

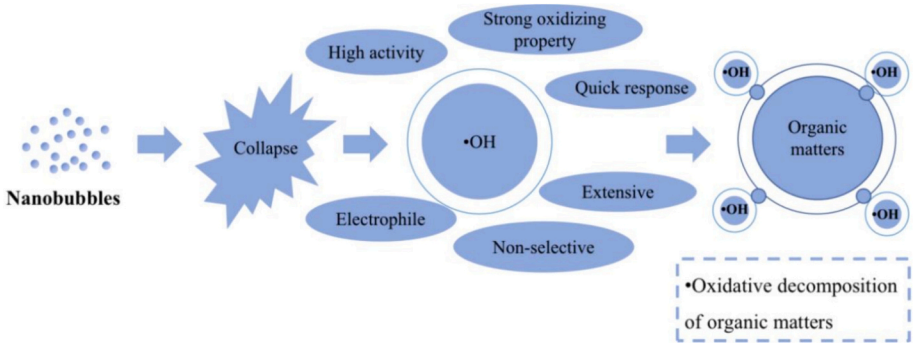


Fig. 1. Potential mechanisms of ROS produced by NBW on AD performance. ROS: reactive oxygen species, NBW: nanobubble water, AD: anaerobic digestion.

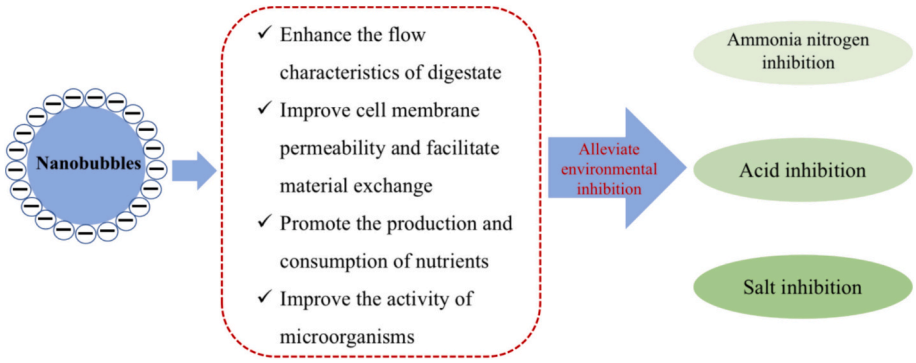


Fig. 2. Addition of NBW to mitigate the inhibition of adverse environmental factors on AD performance. NBW: nanobubble water, AD: anaerobic digestion.

respectively in reactors with 0–30 g NaCl/L upon the addition of air-NBW can be attributed to two possible mechanisms: (1) the addition of air-NBW could improve the electron transfer and corresponding enzyme activity of microorganisms under high salinity, and (2) negatively charged NB would attract Na^+ to form an electric double layer in a high salt environment, which may avoid the imbalance of microbial osmotic pressure caused by excess Na^+ entering the cells (Hou et al., 2021a). For ammonia inhibition, Fan et al. (2021b) performed AD experiments using ammonia-rich pig manure as a substrate and investigated the effect of NBW addition on the stability of ammonia-inhibited reactors. The addition of air-NBW was found to be effective in alleviating the lower methanogenesis rate under ammonia inhibition by stimulating hydrolase and coenzyme F_{420} activity, and methane production was increased by 12 %–39 %. In practical applications, AD may encounter the inhibitory effects of more factors. Therefore, the effectiveness of NBW addition to alleviating the inhibitory effects has considerable potential in practical applications, and it is worthwhile to further investigate the detailed mechanism.

3.3. Improvement of microbial activity

Microorganisms are the core of the whole AD system because the degradation of organic matter and the production of biogas are achieved through various metabolic activities of anaerobic microorganisms. Therefore, the AD process involves a variety of microorganisms (hydrolytic bacteria, acetogens, methanogens, and so on), and the activity of microorganisms has a significant impact on the sustainability and stability of AD systems. Fortunately, most recent studies have demonstrated that NBW as an additive in AD systems could stimulate microbial activity and improve microbial community structure, thereby increasing methane production. The effects of NBW addition on microorganisms in the AD process are displayed in Fig. 3.

For instance, according to research by Yang et al. (2019), the addition of N_2-NBW led to more negatively charged sludge particles in the waste activated sludge, and extracellular hydrolase activities were increased by 14 %–17 % with a 29 % increase in methane production. Meanwhile, it was observed by Fan et al. (2021a)that air-NBW addition could stimulate the activity of hydrolase and coenzyme F_{420}, which is consistent with the results obtained by Hou et al. (2021b). Similarly, another study reported that NBW could increase the activity of coenzyme F_{420} by 1.23 times in the methanogenic stage of AD, greatly improving methane production (Liu, 2018). Otherwise, Wang et al. (2020c) utilized O_2-NBW to improve the methanogenic performance in the AD of a straw-based lignocellulose reactor, in which the cellulase activity was increased by 38 % and the cellulose degradation rate was increased by about 14 %. The reasons why the addition of NBW could improve the activities of various microbial enzymes may be as follows: (1) the high surface charge, zeta potential, and mobility of NB could adsorb organic matter and accelerate the rate of nutrient transfer, indirectly stimulating enzyme activity; (2) NB has a larger specific surface area and contact angle, which could increase the contact area of microorganisms and improve the rate of the biochemical reaction.

Apart from increasing enzyme activity, the addition of NBW could improve the microbial community structure in the AD system and optimize the abundance distribution and metabolic pathways of different functional microorganisms. A growing body of research indicates that the microaerobic environment could strengthen the activity of facultative anaerobic bacteria and promote substrate decomposition, which greatly accelerates the hydrolysis stage during the AD process. It is noteworthy that the release of O_2 from the collapse of air-NBW and O_2-NBW could create the micro-oxygen environment. As shown in Fig. 4 (a), the dissolved oxygen (DO) in O_2-NBW (8.28 mg/L) was higher than

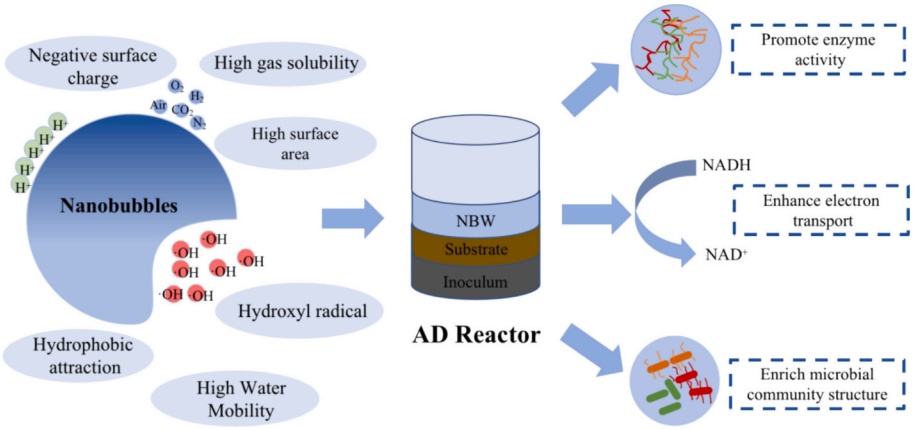


Fig. 3. The effects of NBW addition on microorganisms in AD process. NBW: nanobubble water, AD: anaerobic digestion.

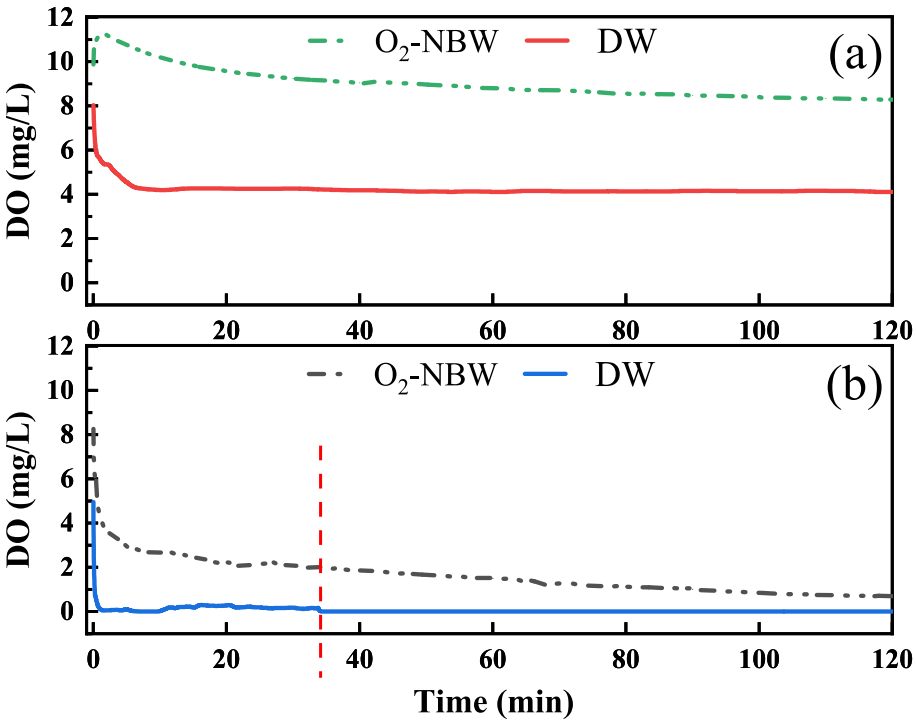


Fig. 4. Changes of DO concentration: (a) in DW (b) in digestate. DO: dissolved oxygen, DW: deionized water.

that in deionized water (DW) (4.13 mg/L). Additionally, the DO in the digestate was depleted after 34 min, and then a strict anaerobic environment was formed. In contrast, the DO in the O_2-NBW added reactor was still maintained at 0.7 mg/L after 2 h (shown in Fig. 4 (b)). Wang et al. (2020c) demonstrated that the micro-oxygen atmosphere generated by the collapse of O_2-NB led to a 1.7-fold increase in the activity of the electron transport system (ETS) in the reactor supplemented with O_2-NBW compared to the control group. Similarly, the research conducted by Hou et al. (2021b) revealed that the micro-oxygen environment created by air-NBW could enhance the hydrolysis acidification stage of FW and did not suppress the methane production during the AD process, but instead facilitated it.

Simultaneously, a higher abundance of the hydrolytic bacteria and acidogens (*Thermotoga, Bacteroidetes, Spirochaetes, and* so on) was detected in the reactors supplemented with air-NBW and O_2-NBW than that in the control reactor (supplemented with DW), especially the highest abundance of *Bacteroides* obtained from the reactors supplemented with O_2-NBW (Wang et al., 2020c). It was also found that the abundance of methanogens (*Methanobacterium* and *Methanomassiliicoccus*) in the reactors supplemented with NBW was higher than that in the control group, which improved the efficiency of the methanogenesis stage (Wang et al., 2020c). These results indicate that the micro-oxygen environment provided by air-NBW and O_2-NBW facilitated the establishment of the direct interspecies electron transfer, enriching microbial diversity in the hydrolytic acidification stage and methanogenesis stage.

For the methanogenic stage, microbial metabolic pathways are primarily categorized into aceticlastic methanogenesis and hydrogenotrophic methanogenesis (Guengerich and Yoshimoto, 2018). On the one hand, the abundance of aceticlastic methanogens in the air-NBW and CO_2-NBW added reactors was higher than that in the control group, with the abundance of aceticlastic methanogens in the air-NBW added reactor (73.4 %) being higher than that in the CO_2-NBW added reactors (52.7 %) (Wang et al., 2020b). This indicates that the addition of NBW could accelerate the aceticlastic methanogenic metabolic pathway by enriching the aceticlastic methanogens. On the other hand, the abundance of hydrogenotrophic methanogens was found to be higher in the CO_2-NBW reactors (0.09 %) than in the air-NBW added reactor (0.04 %), while hydrogenotrophic methanogens were not observed in the control reactor. Up to date, relatively few studies have been conducted on the influence of NB on the microbial community structure and metabolic pathways in the AD system. Microorganisms, as the core throughout the AD process, deserve to be studied intensively in the NBW-based AD system. In addition, H_2 and CO_2 are the substrates of the hydrogenotrophic methanogenesis process, which can upgrade biogas in situ and increase methane production (Bassani et al., 2015; Luo and Angelidaki, 2012; Tian et al., 2019). The effects of H_2-NB and CO_2-NB on microbial metabolic activity and microbial community structure in the hydrogenotrophic methanogenic pathway also should be followed more closely.

3.4. Enhancement of mass transfer efficiency

NBW has a longer spin-spin relaxation time (T_2) when compared to normal water, and the number of NB shows a positive correlation with the T_2 value of water, which represents a higher water molecular mobility (Liu et al., 2013; Ushikubo et al., 2010). Multitudinous previous studies focused on the improvement of microbial activity and AD performance by NBW while ignoring the variation of digestate properties after adding NBW. The AD process of microorganisms is accomplished in the digestate, so the variation of digestate characteristics primarily influences AD performance. The high mobility of NBW may ameliorate the high viscosity of digestate and enhance the fluidity and mass transfer efficiency of digestate. In addition, the improvement in the fluidity of the digestate probably enhances the mass transfer efficiency of nutrients to microorganisms. A recent study has found that NB had high mobility, and the consumption of soluble proteins and carbohydrates was enhanced in the NBW group during the AD process (Fan et al., 2020a). In another study, it was also mentioned that NB could increase the mobility of water, permitting more nutrients to be transferred to microbial cells with faster transfer rates (Fan et al., 2020b).

In general, the mass transfer efficiency of gases depends largely on the mass transfer area of the gas-liquid phase. Compared with CB, NB has a larger gas-liquid contact area and offers a higher gas transfer

efficiency (Zhang et al., 2020; Zhang et al., 2008). Meanwhile, the NB have higher gas solubility in the solution due to their higher internal pressure and could accelerate the gas transfer efficiency in the solution. The efficiency of oxygen transfer in water could be increased by 1.5 times by O_2-NB, which dramatically solved the problem of insufficient supply of dissolved oxygen in water and accelerated the growth of cells (Xiao and Xu, 2020). Furthermore, the research of Cerrón-Calle et al. (2022) determined that K_L•a value for CO_2-NBs was 0.35 h^{-1}, being 11-fold higher than the conventional bubbles (0.03 h^{-1}). These results proved that NB could improve the mass transfer efficiency of digestate and promote the dissolution of gas. Other than that, NB could modify the transport mechanism of gas delivery to the cell membrane or cells, and then influence transmembrane proteins or membrane structures and improve the efficiency of material transfer inside and outside the cell (Seddon et al., 2012). Therefore, the property that NB could improve gas transfer efficiency is probably also instrumental in microbial growth and metabolism during the AD process.

As a conclusion, NB could improve water mobility and gas transfer efficiency and similarly may enhance the various mass transfer characteristics of the digestate. At present, multitudinous studies about NB to improve mass transfer were carried out in aqueous solution. In contrast, various metabolic activities of microorganisms in the AD process were completed in the digestate which has a high viscosity and complex internal environment. Obviously, the characteristics of NB in water cannot fully represent the characteristics and action mechanism of NB in the digestate. Therefore, it is important to highlight the influence of NB on the mass transfer characteristics of digestate to reveal the detailed mechanism of NB to enhance the AD process.

Overall, the addition of NBW to intensify AD of biomass was mainly by facilitating the hydrolysis of substrates and improving the mass transfer efficiency of the digestate and the metabolic functioning system of microorganisms, and the potential mechanisms are shown in Fig. 5.

4. Potentials of NB pretreatment for the enhanced AD process

4.1. Pretreatment on inoculum

The low metabolic activity of microorganisms, especially methanogenic bacteria, inevitably results in a slow start-up and a long operating cycle of the AD reactor. Therefore, enhancement of microbial activity may be an effective method to solve these problems. According to the results of previous studies, NBW addition could remarkably increase methane production by increasing microbial activity and improving microbial community structure. Therefore, Wang et al. (2020a) proposed for the first time that NBW was used to pre-domesticate AD sludge to improve microbial activity and accelerate reactor start-up. This strategy could effectively reduce the volume of the AD reactors when compared to adding NBW directly to the AD reactor. As expected, after the sixth pre-domesticate, the ETS activity of AD sludge was improved significantly. Specifically, the ETS activity was ordered as O_2-NBW > N_2-NBW > DW, which was increased by 20 % and 9 % compared with the DW group. Subsequently, the pre-domesticated sludge was used as inoculum for AD of corn straw, and the results showed that the reactor start-up time was shortened in the AD reactor inoculated with the O_2-NBW and N_2-NBW pre-domesticated sludge. For AD performance, methane production was increased by 22 % and 10 %, and the degradation rates of cellulose and hemicellulose increased by 88 % and 23 % compared with the control group.

In order to improve the problem of low microbial activity in high-salt environments, Hou et al. (2022) used air-NBW to pre-domesticated AD sludge as an inoculum for AD of high salinity FW, and then supplemented K^+ in the methanogenic phase to further weaken salt inhibition. Sludge activity was higher by 27 % in the air-NBW reactor than in the DW reactor after the second domestication cycle, and hydrogen production was increased by 46 % and methane production was increased by 17 %–25 % in the subsequent two-stage AD system. At the same time, the lag time of the air-NBW pre-domesticate sludge reactor was significantly shorter than that of the DW pre-domesticated sludge reactor. These studies suggest that pre-domesticating AD sludge with NBW could be an excellent strategy to advance microbial activity while increasing methane production efficiency without increasing the volume of the AD reactor.

4.2. Pretreatment on substrates

Aeration treatment could accelerate the transfer of oxygen to the liquid and strengthen the full contact between the organic matter and microorganisms, which would facilitate the degradation of organic matter (Åmand et al., 2013; Rosso et al., 2008). Zhang et al. (2019) investigated the effects of air aeration and oxygen aeration under

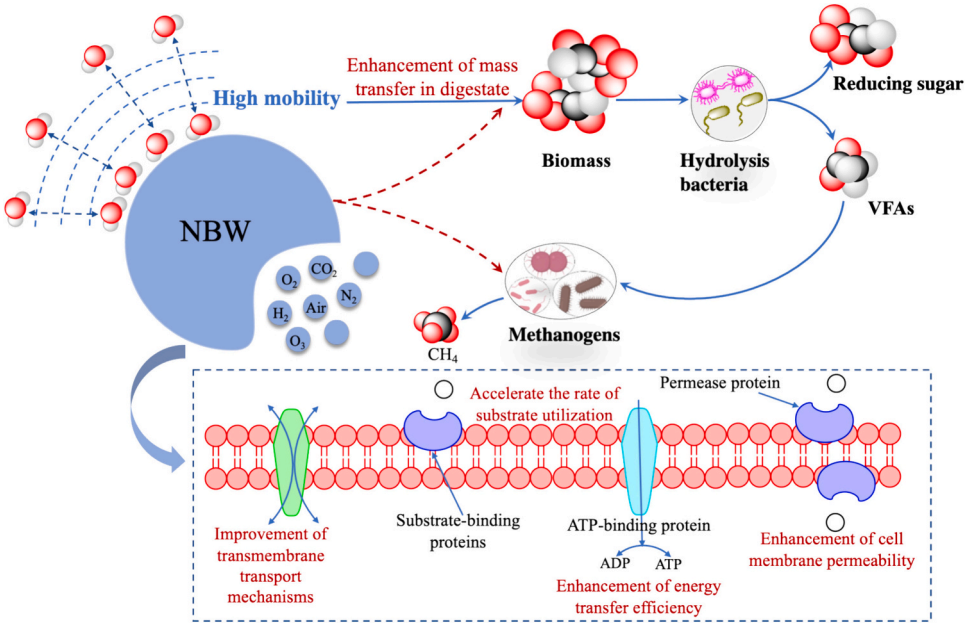


Fig. 5. Potential mechanisms for enhanced AD of biomass by NB technology. AD: anaerobic digestion, NB: nanobubble.

different feed-to-microorganisms ratios (F/M) on the degradation of organic matter and extracellular products in activated sludge. The results found that the degradation rate of organic matter in the reactor with oxygen aeration was much higher than that in the reactors with air aeration and no aeration under the high F/M condition. Meanwhile, the concentration of extracellular polymer was decreased, and the concentration of soluble microbial products was increased significantly after oxygen aeration treatment. More interestingly, the oxygen microbubble aeration treatment not only optimized the community structure of aerobic microorganisms, leading to a 25 % surge in the total relative abundance of the top 10 genera, but also bolstered the treatment efficiency of coal gasification wastewater, achieving 28 % and 36 % higher removal rates of chemical oxygen demand (COD) and total phenols, respectively, compared to the traditional reactor (Zhuang et al., 2016). However, aeration treatment requires specifically complex operations with more energy-consuming and have certain risks. In contrast, NB aeration technology consumes less energy with higher gas solubility during the aeration process. For instance, O_2-NB aeration treatment has been proven to provide more O_2 than conventional aeration treatment, improving air utilization while consuming only 20 % of the aeration energy (Xiao and Xu, 2020). Up to now, there is still an absence of research on the pretreatment of AD substrates using NB aeration. Based on the application of NB aeration technology in other fields, it may provide an innovative approach to the pretreatment of substrates in the AD system.

Undoubtedly, complex organic solid wastes are frequently difficult to be fully utilized by microorganisms during the AD process, therefore, improving the degradation of substrates is a major challenge at present. For instance, lignocellulosic, as a high-complexity organic polymer, is extremely difficult to degrade directly by microorganisms, and lack of oxygen or ROS in an anaerobic environment is one of the main reasons for the difficulty of lignocellulosic degradation (Khan and Ahring, 2019). Guengerich and Yoshimoto (2018) and Waggoner et al. (2015) reported that •OH radicals could induce C—O and C—C sound cleavage in the lignin molecule to depolymerize the lignin and attack the active site on the benzene ring to achieve ring opening. Almomani et al. (2019) directly used •OH radicals generated by advanced oxidation processes (AOPs, Fenton, ozone, and ozone combined with Fe(II) and H_2O_2) to enhance the oxidative decomposition of solid wastes and the subsequent AD process. The results showed that the solids and COD of the substrate were decreased by 29.5 % and 37.5 %, the soluble matter was increased by 6 times, and the cumulative methane production was increased by 30 %. It is noteworthy that one of the key properties of NB is the production of •OH radicals upon collapse, and therefore the reaction of •OH radicals in NBW with lignocellulosic substances is utilized for depolymerizing lignin. Previous studies were conducted to enhance the degradation of lignocellulose by adding NBW directly to AD reactors. For example, it was found that the lignin degradation rate was increased by 10 % and methane yield by 17 % compared to the control group after N_2-NBW was added to the waste activated sludge and alkaline lignin anaerobic co-digestion system (Yang et al., 2020). Ho et al. (2020) compared the effect of N_2-NBW and CO_2-NBW on methane production from lignin mono-digestion and acetate co-digestion. Following 10 days of AD, the reduction rate of lignin was found to reach 19 % and 17 % in the reactors with the addition of N_2-NBW and CO_2-NBW, respectively, while the reduction rate of lignin in the control reactor was only 6 %. These results suggest that the addition of NBW could promote lignin degradation. Therefore, a strategy may be proposed in which NB technology can be used as an approach to pretreat substrates or in combination with other pretreatment methods to improve the degradation of complex substrates. For instance, ultrasonic pretreatment can destroy the complex structure and improve the substrate degradation rate and low-intensity ultrasonic pretreatment can also improve the activity of AD sludge (Tyagi et al., 2019; Zou et al., 2016). Moreover, ultrasound has been found to effectively assist ozone in producing more •OH radicals. As reported by Masuda et al. (2015) and Aluthgun et al. (2020), the collapse of cavitation bubbles caused by ultrasound resulted in the formation of •OH radicals and increasing the power or power intensity led to higher acoustic pressure, enhanced cavitation, and a more violent collapse of bubbles. Therefore, NB technology could be combined with ultrasound for pretreatment of substrates. However, few studies have been conducted on NBW pretreatment substrates, and the feasibility of which needs to be verified.

5. Prospects

The ability of NB technology to improve AD performance and increase methane production could be attributed to the following: (1) the strong oxidative property of •OH radicals produced by the collapse of NB is beneficial for the oxidation of complex organic matter; (2) NB could buffer adverse factors to alleviate environmental inhibition and improve the stability of AD reactors; (3) the high gas solubility of NB allows for a higher gas transfer efficiency, which could affect the transmembrane transport mechanism and speed up the nutrient transfer efficiency; (4) the high water mobility of NB can improve the flow characteristics of solution, thereby increasing nutrient transport efficiency. However, there are still few studies related to enhanced methanogenesis by NB, only limited to AD of substrates such as lignocellulose, FW, animal manure, etc. A detailed mechanism is needed to realize the efficiency and feasibility of NB in practical applications. Future studies may be carried out on the effects of different gas types NBW on the pre-domesticated activated sludge and can also be combined with other pre-domesticated sludge methods. Additionally, the use of NBW as a pretreatment technology for substrates in the AD system has the potential to induce substrate degradation through the production of •OH radicals.

Micro-oxygen pretreatment can also improve the degradation of complex substrates, and thus shorten the AD cycle and increase biogas production (Ali et al., 2018; Jin et al., 2009). O_2-NBW and air-NBW could provide more dissolved oxygen and promote the activity of aerobic and parthenogenic microorganisms in membrane bioreactors (Agarwal et al., 2011; Khan et al., 2020). Therefore, NB technology may be applied to provide aeration in micro-oxygen pretreatment to improve aeration efficiency and reduce energy consumption.

From another perspective, NB has high mobility and gas transfer efficiency, but most of the current studies are derived with water as the matrix. However, in the complicated biochemistry of the AD process, the rising spillage of gases and the transfer of substances within the digestate may have an impact on the bubbles, which causes the actual retention time of NB in the digestate may not be consistent with that in aqueous solution. The conclusions drawn with water as the substrate apparently cannot fully explain the mechanism of AD enhancement by NB technology because the digestate has a higher viscosity and the mass transfer efficiency of digestate is always lower than that of aqueous solution. Further studies should use the actual digestate as the substrate and focus more on the interaction between NB and the digestate, including the Zeta potential, bubbles size and concentrations, digestate mobility, and mass transfer characteristics. This may be crucial in revealing the detailed mechanism of NB to enhance methane production from the AD system. On the other hand, the application of NB technology in the AD field is mostly on a laboratory scale, but this does not provide a basis for the large-scale application of NB technology in practical engineering. Therefore, semi-continuous and continuous experiments should be performed to prove its feasibility in practical applications. In addition, the influence of different gas types of NB on each stage of anaerobic digestion should be noted, and different NB should be used in different stages of AD to form a set of process flows to achieve optimal performance.

6. Conclusions

NB technology was proven to be effective in AD by alleviating

environmental inhibition, enhancing microbial activity, and improving mass transfer efficiency. These factors significantly promote substrate degradation, ameliorate digestate characteristics, and enhance methane production, making NB technology a valuable tool for improving the efficiency of the AD process. More precisely, the •OH generated by the collapse of the NB could facilitate the oxidation of complex organic matter and the degradation of substrates. Future studies should emphasize the impact of NB on anaerobic microorganisms, as well as heat and mass transfer of digestate to uncover the microscopic mechanisms at play within the AD systems. Expanding the application of NB technology in the AD field should involve exploring novel modalities such as pre-treatment of substrates. Simultaneously, with the consideration of economy and rationality, it is vital to optimize the structural parameters of the NB generator to facilitate the application of NB in actual AD projects.

CRediT authorship contribution statement

Tingting Hou: Writing – original draft, Investigation, Data curation, Conceptualization. **Hao Song:** Investigation, Data curation. **Zhiqiang Cui:** Investigation, Data curation. **Chao He:** Writing – review & editing, Project administration. **Liang Liu:** Writing – review & editing. **Pengfei Li:** Writing – review & editing. **Gang Li:** Writing – review & editing. **Quanguo Zhang:** Writing – review & editing. **Zhenya Zhang:** Writing – review & editing. **Zhongfang Lei:** Writing – review & editing. **Yuri V. Litti:** Writing – review & editing. **Youzhou Jiao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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